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Segurança em Sistemas Operacionais e Redes - II

Reconhecimento – NC e NMAP

Reconhecer e Enumerar

- **Para enumerar os serviços iremos utilizar alguns recursos, o primeiro é o NetCat – NC**

- **#nc www.bancocn.com 80**
  - Agora conseguimos enviar dados.
  - Retornou um Bad Request, porque o dados que enviamos não são dados de conexão HTTP.
  - Se enviar de forma correta, conforme o exemplo abaixo, teremos a resposta de retorno do acesso.
    - GET / HTTP/1.1
    - Host: www.bancocn.com

Reconhecer e Enumerar

```
File  Actions  Edit  View  Help
</html>

┌──(kali㉿kali)-[~]
└─$ nc www.bancocn.com 80
GET / HTTP/1.1
Host: www.bancocn.com

HTTP/1.1 403 Forbidden
Date: Mon, 28 Feb 2022 03:16:26 GMT
Content-Type: text/html; charset=UTF-8
Transfer-Encoding: chunked
Connection: close
Permissions-Policy: accelerometer=(),autoplay=(),camera=(),clipboard-read=
(),clipboard-write=(),fullscreen=(),geolocation=(),gyroscope=(),hid=(),int
erest-cohort=(),magnetometer=(),microphone=(),payment=(),publickey-credent
ials-get=(),screen-wake-lock=(),serial=(),sync-xhr=(),usb=()
Cache-Control: private, max-age=0, no-store, no-cache, must-revalidate, po
st-check=0, pre-check=0
Expires: Thu, 01 Jan 1970 00:00:01 GMT
X-Frame-Options: SAMEORIGIN
Report-To: {"endpoints":[{"url":"https:\/\/a.nel.cloudflare.com\/report\/v
```

Reconhecer e Enumerar

- Para porta que não tem serviço, a resposta não existe.

Port Scanner

- Com o NMAP, temos um retorno mais direto:

```
kali@kali: ~
File  Actions  Edit  View  Help

┌──(kali㉿kali)-[~]
└─$ nmap www.bancocn.com
Starting Nmap 7.92 ( https://nmap.org ) at 2022-02-27 22:38 EST
Nmap scan report for www.bancocn.com (172.67.192.199)
Host is up (0.056s latency).
Other addresses for www.bancocn.com (not scanned): 104.21.52.8 2606:4700:3
034::ac43:c0c7 2606:4700:3033::6815:3408
Not shown: 996 filtered tcp ports (no-response)
PORT     STATE SERVICE
80/tcp   open  http
443/tcp  open  https
8080/tcp open  http-proxy
8443/tcp open  https-alt

Nmap done: 1 IP address (1 host up) scanned in 5.25 seconds

┌──(kali㉿kali)-[~]
└─$
```

Port Scanner

- Scan escolhendo uma porta:

```
File  Actions  Edit  View  Help

┌──(kali㉿kali)-[~]
└─$ nmap bancocn.com -p 80
Starting Nmap 7.92 ( https://nmap.org ) at 2022-02-28 00:01 EST
Nmap scan report for bancocn.com (104.21.52.8)
Host is up (0.053s latency).
Other addresses for bancocn.com (not scanned): 2606:4700:3033::6815:3408 2
606:4700:3034::ac43:c0c7 172.67.192.199

PORT   STATE SERVICE
80/tcp open  http

Nmap done: 1 IP address (1 host up) scanned in 0.21 seconds

┌──(kali㉿kali)-[~]
└─$
```

Port Scanner

- Scan de várias portas simultaneamente:

```
File  Actions  Edit  View  Help

┌──(kali㉿kali)-[~]
└─$ nmap bancocn.com -p 80,21,22,25,8080,3306
Starting Nmap 7.92 ( https://nmap.org ) at 2022-02-28 00:01 EST
Nmap scan report for bancocn.com (172.67.192.199)
Host is up (0.054s latency).
Other addresses for bancocn.com (not scanned): 2606:4700:3033::6815:3408 2
606:4700:3034::ac43:c0c7 104.21.52.8

PORT     STATE    SERVICE
21/tcp   filtered ftp
22/tcp   filtered ssh
25/tcp   filtered smtp
80/tcp   open     http
3306/tcp filtered mysql
8080/tcp open     http-proxy

Nmap done: 1 IP address (1 host up) scanned in 1.57 seconds

┌──(kali㉿kali)-[~]
└─$
```

Port Scanner

- Detecção dos Serviços:

```
kali@kali: ~
File  Actions  Edit  View  Help
└─$ nmap bancocn.com -p 80,21,22,25,8080,3306 -sV
Starting Nmap 7.92 ( https://nmap.org ) at 2022-02-28 00:02 EST
Nmap scan report for bancocn.com (104.21.52.8)
Host is up (0.052s latency).
Other addresses for bancocn.com (not scanned): 2606:4700:3033::6815:3408 2
606:4700:3034::ac43:c0c7 172.67.192.199

PORT     STATE    SERVICE VERSION
21/tcp   filtered ftp
22/tcp   filtered ssh
25/tcp   filtered smtp
80/tcp   open     http    Cloudflare http proxy
3306/tcp filtered mysql
8080/tcp open     http    Cloudflare http proxy

Service detection performed. Please report any incorrect results at https:
//nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 8.63 seconds

┌──(kali㉿kali)-[~]
└─$
```

Port Scanner

- Descoberta de hosts:

```
┌──(kali㉿kali)-[~]
└─$ nmap 104.21.52.0-100 -sn
Starting Nmap 7.92 ( https://nmap.org ) at 2022-02-28 00:07 EST
```

```
File  Actions  Edit  View  Help
Host is up (0.053s latency).
Nmap scan report for 104.21.52.93
Host is up (0.062s latency).
Nmap scan report for 104.21.52.94
Host is up (0.062s latency).
Nmap scan report for 104.21.52.95
Host is up (0.057s latency).
Nmap scan report for 104.21.52.96
Host is up (0.057s latency).
Nmap scan report for 104.21.52.97
Host is up (0.063s latency).
Nmap scan report for 104.21.52.98
Host is up (0.054s latency).
Nmap scan report for 104.21.52.99
Host is up (0.056s latency).
Nmap scan report for 104.21.52.100
Host is up (0.060s latency).
Nmap done: 101 IP addresses (101 hosts up) scanned in 0.55 seconds

┌──(kali㉿kali)-[~]
└─$
```

Port Scanner

- Ping simples com temporização:

```
┌──(kali㉿kali)-[~]
└─$ nmap 104.16.0.0/12 -sn -T5
Starting Nmap 7.92 ( https://nmap.org ) at 2022-02-28 00:16 EST
Stats: 0:00:01 elapsed; 0 hosts completed (0 up), 4096 undergoing Ping Sca
n
Ping Scan Timing: About 14.76% done; ETC: 00:16 (0:00:06 remaining)
Stats: 0:00:01 elapsed; 0 hosts completed (0 up), 4096 undergoing Ping Sca
n
Ping Scan Timing: About 27.40% done; ETC: 00:16 (0:00:05 remaining)
```

Port Scanner

- Ping Simples:

```
File  Actions  Edit  View  Help

┌──(kali㉿kali)-[~]
└─$ nmap 192.168.1.0-255 -sn
Starting Nmap 7.92 ( https://nmap.org ) at 2022-02-28 09:17 EST
Nmap scan report for 192.168.1.2
Host is up (0.0025s latency).
Nmap scan report for www.webgui.Nokiawifi.com (192.168.1.254)
Host is up (0.0078s latency).
Nmap done: 256 IP addresses (2 hosts up) scanned in 2.68 seconds

┌──(kali㉿kali)-[~]
└─$
```

Port Scanner

- Diferentes Técnicas:

```
SCAN TECHNIQUES:
  -sS/sT/sA/sW/sM: TCP SYN/Connect()/ACK/Window/Maimon scans
  -sU: UDP Scan
  -sN/sF/sX: TCP Null, FIN, and Xmas scans
  --scanflags <flags>: Customize TCP scan flags
  -sI <zombie host[:probeport]>: Idle scan
  -sY/sZ: SCTP INIT/COOKIE-ECHO scans
  -sO: IP protocol scan
  -b <FTP relay host>: FTP bounce scan
PORT SPECIFICATION AND SCAN ORDER:
```

Port Scanner

- Reconhecer o Sistema Operacional:

```
File  Actions  Edit  View  Help
┌──(kali㉿kali)-[~]
└─$ sudo nmap -O 192.168.1.16
[sudo] password for kali:
Starting Nmap 7.92 ( https://nmap.org ) at 2022-02-28 09:50 EST
Stats: 0:00:01 elapsed; 0 hosts completed (0 up), 1 undergoing ARP Ping Sca
n
Parallel DNS resolution of 1 host. Timing: About 0.00% done
Nmap scan report for 192.168.1.16
Host is up (0.018s latency).
All 1000 scanned ports on 192.168.1.16 are in ignored states.
Not shown: 1000 closed tcp ports (reset)
MAC Address: 6E:EC:80:13:91:24 (Unknown)
Too many fingerprints match this host to give specific OS details
Network Distance: 1 hop

OS detection performed. Please report any incorrect results at https://nmap
.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 33.06 seconds

┌──(kali㉿kali)-[~]
└─$
```

Port Scanner

- Modo Stealth e reconhecimento dos serviços:

```
┌──(kali㉿kali)-[~]
└─$ sudo nmap 192.168.1.254 -sS -sV                                  1 ✗
Starting Nmap 7.92 ( https://nmap.org ) at 2022-02-28 09:58 EST
```

Port Scanner

- Modo Stealth e reconhecimento dos serviços:

```
File  Actions  Edit  View  Help
Service scan Timing: About 66.67% done; ETC: 10:00 (0:00:11 remaining)
WARNING: Service 192.168.1.254:443 had already soft-matched rtsp, but now s
oft-matched sip; ignoring second value
Nmap scan report for www.webgui.Nokiawifi.com (192.168.1.254)
Host is up (0.042s latency).
Not shown: 995 closed tcp ports (reset)
PORT     STATE    SERVICE  VERSION
22/tcp   filtered ssh
23/tcp   filtered telnet
80/tcp   open     rtsp
443/tcp  open     ssl/rtsp
8099/tcp open     unknown
3 services unrecognized despite returning data. If you know the service/ver
sion, please submit the following fingerprints at https://nmap.org/cgi-bin/
submit.cgi?new-service :
==============NEXT SERVICE FINGERPRINT (SUBMIT INDIVIDUALLY)==============
SF-Port80-TCP:V=7.92%I=7%D=2/28%Time=621CE35E%P=x86_64-pc-linux-gnu%r(GetR
SF:equest,1109,"HTTP/1\.0\x20200\x20OK\r\nCache-Control:private,max-age=0;
SF:\r\nX-Frame-Options:SAMEORIGIN\r\nX-Content-Type-Options:nosniff\r\nX-X
SF:SS-Protection:1;\x20mode=block\r\nContent-Security-Policy:default-src\x
SF:20'self'\x20blob:\x20'unsafe-inline'\x20'unsafe-eval'\r\nStrict-Transpo
```

Port Scanner

- Com o –A temos:
  - Detecção de sistema operacional (-O)
  - Detecção de versão dos serviços (-sV)
  - Scan de scripts padrão do Nmap (--script=default)
  - Traceroute (se possível)

```
┌──(kali㉿kali)-[~]
└─$ nmap -A bancocn.com                                          255 ✖
Starting Nmap 7.92 ( https://nmap.org ) at 2022-02-28 10:01 EST
Nmap scan report for bancocn.com (172.67.192.199)
Host is up (0.054s latency).
Other addresses for bancocn.com (not scanned): 2606:4700:3034::ac43:c0c7 26
06:4700:3033::6815:3408 104.21.52.8
Not shown: 996 filtered tcp ports (no-response)
PORT     STATE SERVICE        VERSION
80/tcp   open  http           Cloudflare http proxy
|_http-title: Did not follow redirect to http://www.bancocn.com/
|_http-server-header: cloudflare
443/tcp  open  ssl/https      cloudflare
```

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